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SAHYADRI

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MANGALURU

Assignment

MongoDB

(CS62298CD)

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Under the Guidance of

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QR Code and URL	Max Marks	Marks Obtained
https://food-rush-wine.vercel.app/	20	

ABSTRACT

The Online Food Ordering System is a web-based application designed to simplify and enhance the process of ordering food from restaurants through a digital platform. The system enables users to browse restaurants, explore menu items, add products to a cart, and place orders efficiently. The project is developed using the MERN stack, which includes MongoDB, Express.js, React.js, and Node.js. MongoDB is used as the database to store and manage restaurant details, customer information, menu items, and order records using a flexible document-oriented data model. The application also provides features such as user authentication, restaurant search, cart management, and order tracking to improve user experience and operational efficiency. The main objective of this project is to provide a scalable, fast, and user-friendly food ordering platform while overcoming the limitations of traditional ordering methods. This project demonstrates the practical use of MongoDB in managing dynamic and real-time web application data.

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Chapter 1

INTRODUCTION

The rapid advancement of internet technologies and digital platforms has significantly transformed the food service industry. Online food ordering systems have emerged as an essential solution for restaurants and customers by enabling convenient and efficient food ordering through web and mobile applications. These systems allow users to browse restaurant menus, place orders, and track deliveries without the limitations of traditional phone-based ordering methods. The increasing popularity of online food delivery applications has improved customer convenience, reduced waiting time, and enhanced overall user satisfaction in the food service sector [1][2].

Traditional food ordering systems often face challenges such as inaccurate order processing, communication delays, limited menu accessibility, and inefficient order management. To overcome these issues, modern web-based applications use scalable technologies and database systems that can efficiently manage large volumes of dynamic data. Research on scalable food ordering platforms using the MERN stack highlights the importance of integrating modern web technologies to improve system performance, scalability, and user experience [3]. Recent studies also demonstrate the effectiveness of MERN stack applications in developing responsive and efficient food delivery systems with advanced functionalities [6][7].

Database management plays a critical role in online food ordering applications because the system continuously handles customer records, restaurant details, menu items, and order transactions. Traditional relational databases may face limitations in handling highly dynamic and rapidly changing application data structures. As a result, NoSQL databases such as MongoDB have gained popularity due to their flexibility, scalability, and high-performance data management capabilities [4]. MongoDB uses a document-oriented model that stores data in JSON-like documents, making it suitable for managing nested and dynamic information such as restaurant menus and customer orders [5].

In this project, an Online Food Ordering System named “FoodRush” is developed using the MERN stack, which includes MongoDB, Express.js, React.js, and Node.js. MongoDB is used as the primary database for storing and managing restaurant information, menu details, customer profiles, and order records efficiently. The application provides functionalities such as user authentication, restaurant browsing, cart management, and order tracking. JWT-based authentication mechanisms are also incorporated to ensure secure access to user accounts and protected resources [8].

This project demonstrates the practical implementation of MongoDB in a real-world web application and highlights the advantages of NoSQL databases in building scalable and flexible online food ordering systems. The system aims to provide an efficient, user-friendly, and reliable platform that improves the overall food ordering experience for both customers and restaurants.

Chapter 2

PROBLEM FORMULATION

2.1 Problem Statement

Traditional food ordering methods often face issues such as delayed communication, incorrect order processing, limited menu accessibility, and inefficient order management. Many existing systems also struggle to manage dynamic restaurant menus and large volumes of customer and order data efficiently. Traditional relational databases may not provide enough flexibility for handling rapidly changing and nested data structures used in modern food ordering applications.

Therefore, there is a need for a scalable and efficient Online Food Ordering System that provides secure user authentication, efficient order management, and flexible data handling using MongoDB and the MERN stack.

2.2 Problem Description

With the increasing demand for online food delivery services, customers expect fast and reliable platforms for browsing menus and placing orders. However, traditional ordering systems are often inefficient and may lead to communication errors, delayed confirmations, and poor customer experience.

Managing restaurant menus, customer records, and order details becomes challenging when the system handles large amounts of dynamic data. Traditional database systems may face difficulties in storing flexible and continuously changing information such as menu updates and customer orders.

To solve these issues, the proposed Online Food Ordering System uses MongoDB, a NoSQL database that efficiently manages dynamic and document-based data. The system is developed using the MERN stack to provide features such as restaurant browsing, cart management, secure authentication, and order tracking. The proposed solution improves scalability, flexibility, and overall user experience in online food ordering services.

2.3 Objectives of Project

The primary objectives of the Online Food Ordering System are as follows:

1. To design and develop a scalable web-based Online Food Ordering System using the MERN stack and MongoDB for efficient data management.
2. To provide users with features such as restaurant browsing, menu exploration, cart management, secure user authentication, and online order placement.
3. To utilize MongoDB’s document-oriented database model for storing and managing restaurant details, customer information, menu items, and order records efficiently.
4. To improve the efficiency and reliability of food ordering services by reducing communication delays and manual order management issues.
5. To develop a user-friendly and responsive platform that enhances customer experience and supports efficient order tracking and management.

Chapter 3

METHODOLOGY AND IMPLEMENTATION

The Online Food Ordering System is developed using the MERN stack, which includes MongoDB, Express.js, React.js, and Node.js. The system follows a client-server architecture where the frontend communicates with the backend through REST APIs, and the backend interacts with the MongoDB database for data storage and retrieval. MongoDB is used as the primary database because of its flexibility in handling dynamic and document-oriented data structures such as restaurant menus and customer orders

System Methodology

The methodology of the proposed system consists of the following stages:

1. User Registration and Authentication
   Users can create accounts and log in securely using JWT-based authentication mechanisms.
2. Restaurant and Menu Management
   Restaurant details and menu items are stored in MongoDB collections using document-based schemas.
3. Food Ordering Process
   Customers can browse restaurants, select food items, add them to the cart, and place orders through the application interface.
4. Order Management
   Orders are stored in the database along with customer details, ordered items, total amount, and order status.
5. Order Tracking and History
   Users can view previous orders and track the status of placed orders.

System Architecture

The system follows a three-layer architecture:

- The frontend provides the user interface for interaction.
- The backend handles API requests, authentication, and business logic.
- MongoDB stores restaurant details, customer data, menu items, and order records.

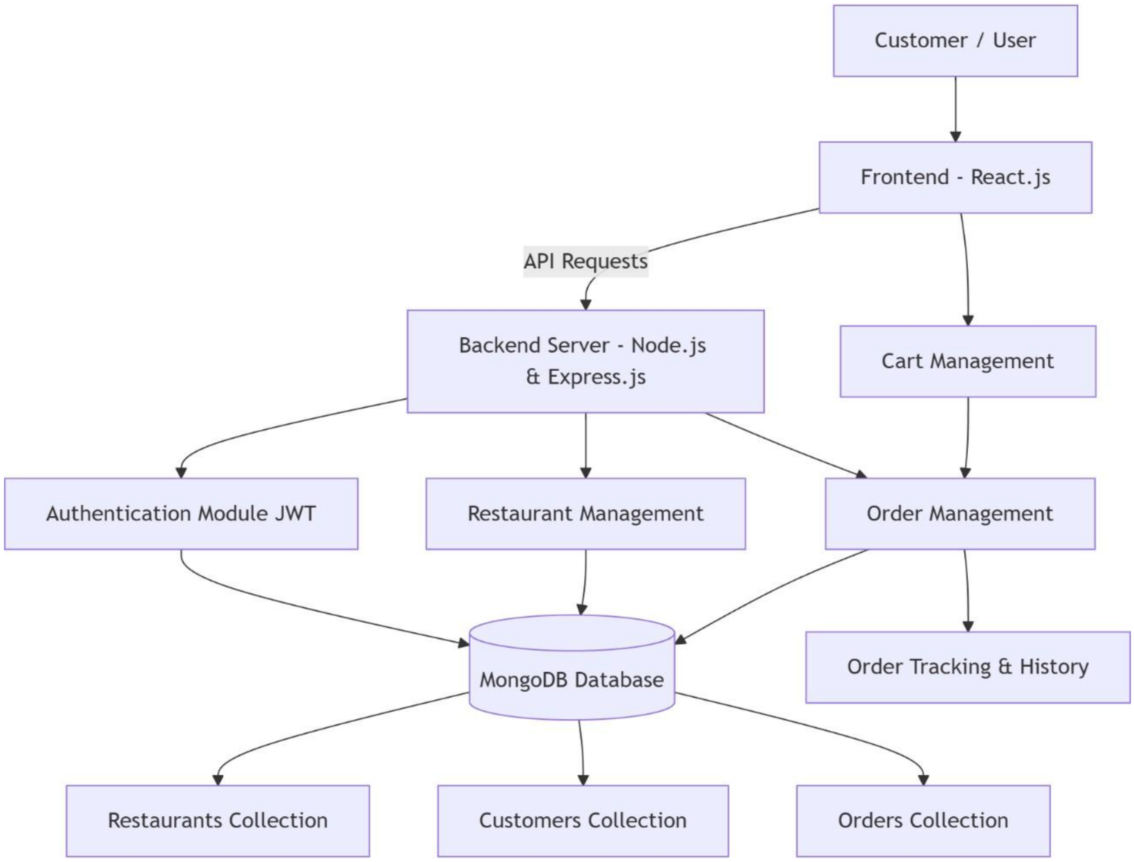

Fig 3.1 : Archititure diagram

Implementation Details

Database Models

Model	Key Fields
Restaurant	name, location, cuisine, menu (Array of menu items)
Customer	username, email, password (hashed using bcryptjs), address
Order	customer_id (ref: Customer), restaurant_id (ref: Restaurant), items (Array), total_amount, status, order_date

Table 3.1: Configuration Parameters for Implementation

Backend Implementation

Server Entry Point (server.js)

The backend server initializes an Express application, connects to MongoDB using the MONGODB_URI stored in the .env file, applies middleware such as CORS and JSON parsing, and registers all API route modules under the /api prefix. The server runs on port 5000 by default.

Route Modules

- authRoutes.js — Handles user authentication including registration and login. Passwords are hashed using bcryptjs and JWT tokens are generated for secure authentication.
- restaurantRoutes.js — Handles fetching restaurant details and menu information.
- orderRoutes.js — Handles order creation, order retrieval, and order status management.

Order Management Process

When a user places an order, the backend validates the request data and stores the order details in the MongoDB database. The order document includes customer information, selected food items, total amount, and order status. The backend then returns a confirmation response to the frontend.

Frontend Implementation

The React frontend is organized into pages and reusable components. React Router is used for navigation between application pages.

Key Pages

- Home.js — Displays restaurant listings with search functionality.
- RestaurantMenu.js — Displays restaurant menu items and allows users to add food items to the cart.
- Cart.js — Displays selected food items and total order amount before order placement.
- Orders.js — Displays user order history and order status tracking.
- Login.js / Signup.js — Handles user authentication and JWT token storage.

Key Components

- Navbar — Provides navigation links and authentication-based rendering.
- RestaurantCard — Displays restaurant information.
- CartComponent — Manages selected food items and quantity updates.
- ProtectedRoute — Restricts access to authenticated users only.

API Endpoints

Endpoints	Description
POST /api/auth/register	Register a new user
POST /api/auth/login	Authenticate user and return JWT
GET /api/restaurants	Retrieve all restaurants
GET /api/restaurants/	Retrieve restaurant details and menu
POST /api/orders	Create a new order
GET /api/orders/	Retrieve order history for a user

Table 3.2 REST API Endpoints

Chapter 4

Project Result

Knowledge Gained

Through the development of the Online Food Ordering System, knowledge was gained in modern web application development using the MERN stack technologies such as MongoDB, Express.js, React.js, and Node.js. The project provided an understanding of NoSQL databases, document-oriented data storage, REST API development, user authentication using JWT, and frontend-backend integration. Knowledge regarding scalable database management and handling dynamic data using MongoDB was also acquired.

Exposure to Technical Literature

The project helped in understanding and analyzing technical papers, journals, and online research resources related to online food ordering systems, MongoDB, and MERN stack application development. It improved the ability to identify credible technical sources and understand modern web development methodologies and database technologies.

Technical Writing Skills

The preparation of this report improved technical writing skills, including structuring content, writing abstracts, documenting methodologies, and properly citing references in academic format. The project also enhanced understanding of maintaining clarity and academic integrity while preparing technical documentation.

Communication Skills

The project improved communication and presentation skills through discussions, project demonstrations, and explanation of technical concepts related to database management, API development, and system architecture. It also enhanced the ability to present technical information using reports and presentation slides.

Technical Skill Development

Practical skills were developed in frontend and backend web development, MongoDB database management, API testing using Postman, and handling project dependencies using Node.js and npm. The project also provided experience in debugging application errors and configuring database connectivity.

Awareness of Sustainable Development Goals (SDGs)

The project contributes to digital transformation and technological innovation in the food service industry, which is connected to Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure). The system promotes efficient digital services and reduces dependency on manual ordering processes, contributing to improved operational efficiency and customer convenience.

Contribution Towards SDGs

The developed Online Food Ordering System supports innovation and digital infrastructure by providing an efficient and scalable platform for online food ordering services. Future improvements such as optimized delivery management and reduced paper-based processes can further contribute toward sustainable and technology-driven solutions in the food service sector.

Figure3: Login Page

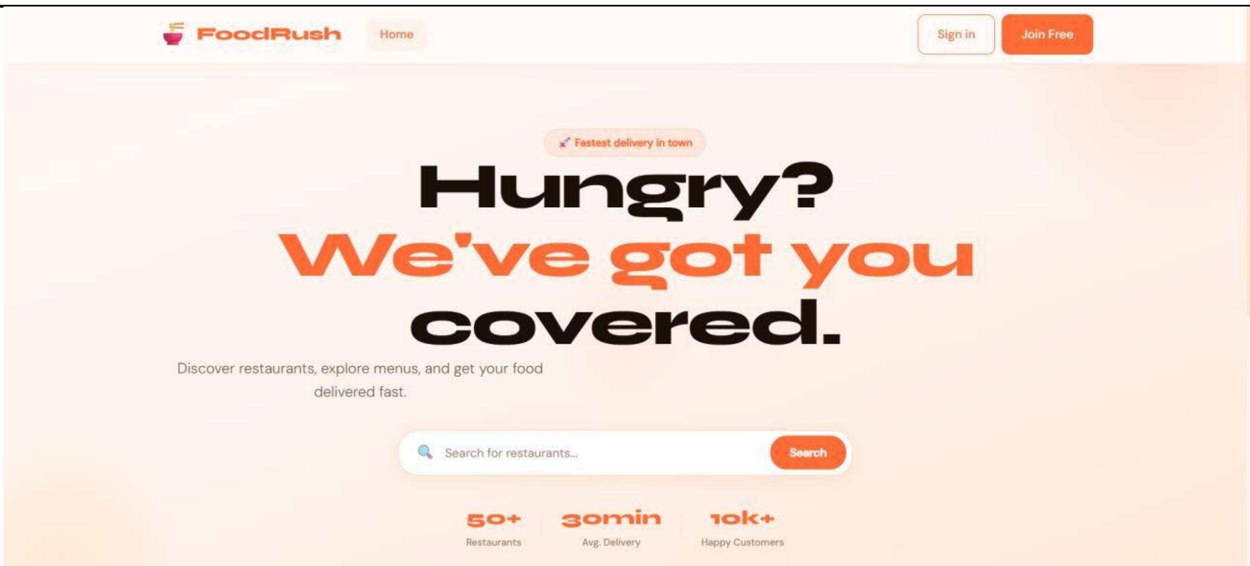

Figure 2 : Home page

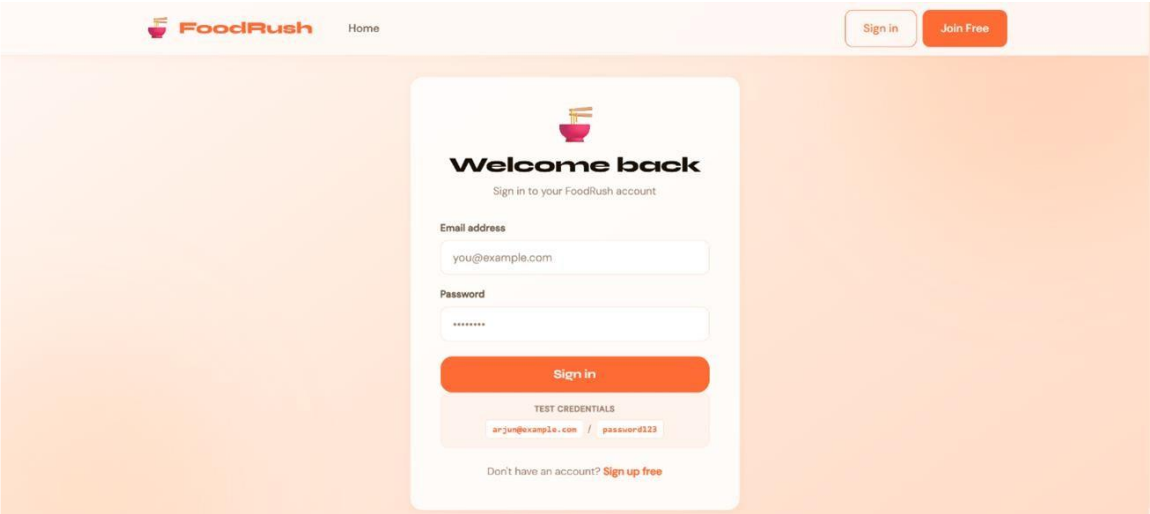

Figure 3 : Login Page

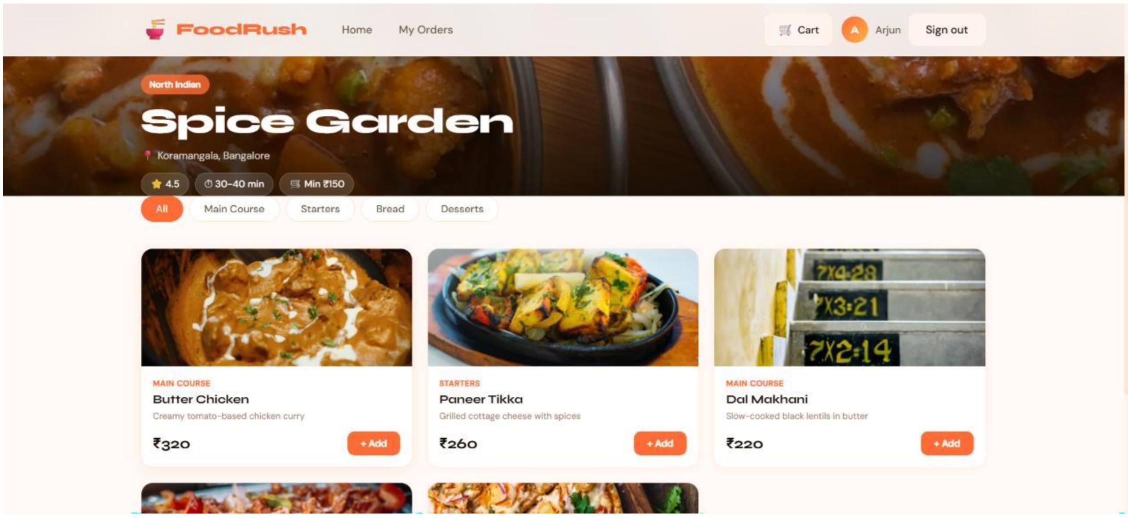

Figure : 4 : Menu

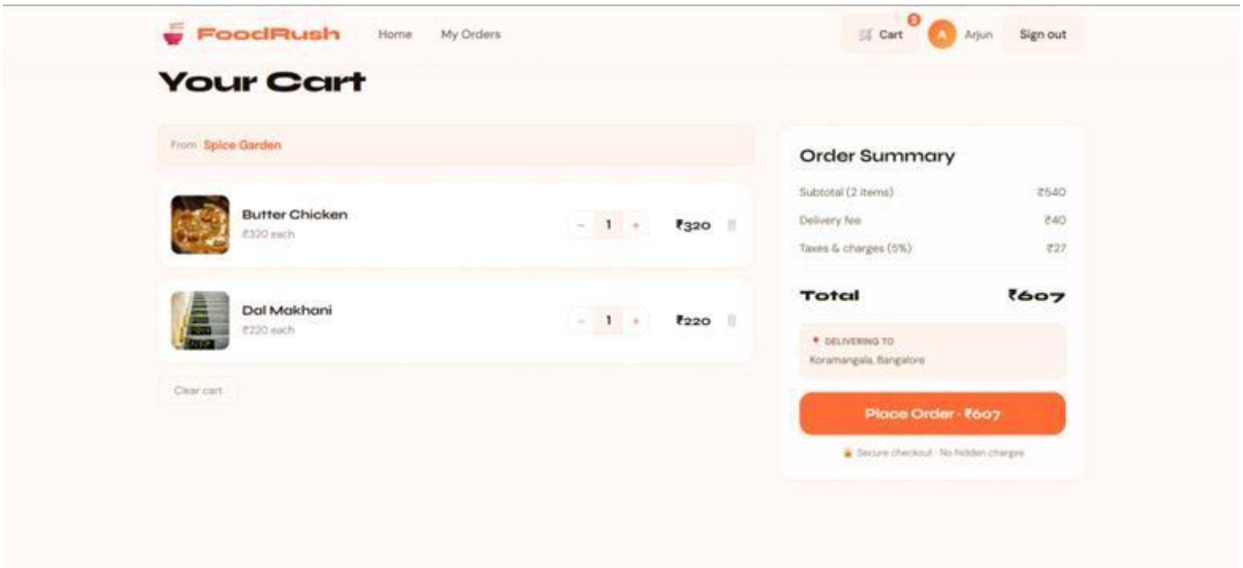

Figure 5 : Shopping cart

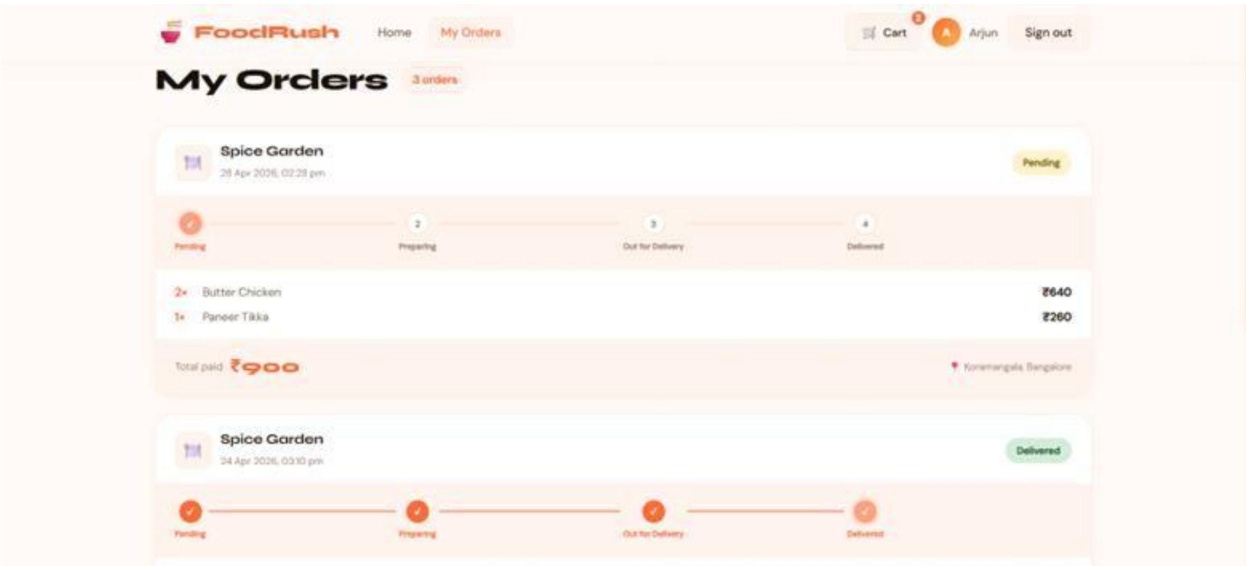

Figure 6 : order tracking

Chapter 5

CONCLUSION

In conclusion, the development of the Online Food Ordering System using the MERN stack and MongoDB demonstrates the effectiveness of modern web technologies in creating scalable and efficient digital platforms for the food service industry. Traditional food ordering methods often face challenges such as delayed communication, inaccurate order management, and limited accessibility. The proposed system addresses these issues by providing a user-friendly platform for restaurant browsing, menu management, online ordering, and order tracking.

The use of MongoDB as a NoSQL database provides flexibility in handling dynamic and document-oriented data such as restaurant menus, customer records, and order details. The integration of React.js, Node.js, and Express.js further enables efficient frontend-backend communication and responsive user interaction.

Through this project, valuable knowledge was gained in web application development, database management, REST API implementation, and secure authentication techniques. The project also highlighted the importance of scalable database solutions and modern web technologies in developing real-world applications.

Overall, the Online Food Ordering System provides a reliable and efficient solution for digital food ordering services while demonstrating the practical application of MongoDB and MERN stack technologies in modern web development.

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Appendix A

APPENDICES

Course Outcome

- CO 1:Identify the research papers/applied knowledge resources on latest trends in area of interest and formulate objectives of the study.
- CO 2:Acquaint literatures review methods and identify the significant technical information relevant to selected topic.
- CO 3:Interpret the observations with hypothesis and summarize the conclusions.
- CO 4:Make use of a logical thought process to efficiently sift findings and produce a well-structured, tailored report.
- CO 5:Develop a detailed presentation of observational outcomes and recommendations for improving future scope.